

Unit-4(Testing of hypothesis)

1. CRITICAL VALUES (MUST REMEMBER)

1.96 → 5% two-tailed
1.645 → 5% one-tailed
2.33 → 1% one-tailed
2.58 → 1% two-tailed

2. TEST IDENTIFICATION (FAST)

Mean + $n \geq 30$ → Z-test
Mean + $n < 30$ → t-test
Proportion → Z-test
Two means → Z / t
Paired data → Paired t-test
Frequencies → Chi-square
Variance → F-test

3. WHICH TAIL?

“>” → Right tail
“<” → Left tail
“≠ / whether / difference” → Two-tail

4 DECISION RULE (WRITE SAME LINE ALWAYS)

If calculated > table → Reject H_0
Else → Do not reject H_0

5. KEY WORD TRICKS

- “Majority” → **P > 0.5**
- “Same population” → **$\mu_1 = \mu_2$**
- “Difference” → **Two-tailed**
- “Increase” → **Right-tailed**
- “Decrease” → **Left-tailed**

FORMULAS

Mean (Z)

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Mean (t)

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Proportion

$$Z = \frac{p - P}{\sqrt{P(1 - P)/n}}$$

Two Proportions

$$Z = \frac{p_1 - p_2}{\sqrt{PQ(1/n_1 + 1/n_2)}}$$

Two Means (t)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

Paired t

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Chi-square

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

F-test

$$F = \frac{\text{larger variance}}{\text{smaller variance}}$$